INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

STRAINERS, SEDIMENT, PIPELINE, SELF-CLEANING (MANUAL OR AUTOMATED)

The General Service Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers both manual and automated self-cleaning pipeline sediment strainers.

1.1 Intended use. The strainers specified are intended for installation in seawater piping systems on the discharge side of the pump.

2. CLASSIFICATION. Self-cleaning strainers shall be of the following types, classes, styles, and sizes as specified (see 7.1).

2.1 Type.

a. Type I - Manual hand wheel actuation.
b. Type II - Electric motor actuation.

2.2 Class.

a. Class 150 - 150 pounds per square inch gage (psig) pressure rating.
b. Class 250 - 250 psig pressure rating.

2.3 Style.

a. Style 1 - One screen, angled flow, Sizes 2, 2½, and 3 inches.
b. Style 2 - Two screens, inline flow, Sizes 2½, 3, 4, 6, 8, 10, and 12 inches.
c. Style 3 - Four screens, inline flow, Sizes 14 and 16 inches.
d. Style 4 - Six screens, inline flow, Size 20 inches.

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3. SALIENT CHARACTERISTICS.

3.1 Codes and standards. Strainers shall comply with the applicable requirements of table I.

Table I. Reference data.

Equipment/Characteristic	Description
Fluid type	Seawater
Operation pressure	150 or 250 psi
Temperature range	70 to 120 °F
Piping connections: Inlet, outlet, and sump port	 150 and 250 rating MIL-PRF-20042 full face flanges
Pressure resistance	FCI 78-1
Flow capacity	Table III in accordance with ISA-75.02
Shock resistance	MIL-S-901, Grade A
Type II strainer only: Motor Control panel	 MIL-M-17059 MIL-DTL-2212

3.2 Design and construction. Each strainer shall be of the type, class, style, and size as specified (see 2.12.1 through 2.3). Each strainer shall contain one or more rotatable screens and fixed scraper blades for the purpose of cleaning the screen(s) without shutting down operation of the system. Compartment covers shall be flanged and attached to the body by bolts. Each cover shall effect a positive seal on an O-ring recessed in the cover plate. Blade inspection/adjustment covers on strainer sizes 4 inches and above shall be bolted to flanges on the strainer body. An O-ring recessed in the cover shall be provided to form a watertight seal. The sump below the screen(s) shall be fitted with a drain outlet.

3.3 Material. The material requirements for all strainer components shall be in accordance with table II.

Table II. Materials required for strainers.

Part	Material	Applicable Documents	Remarks
Bodies, covers (parts in contact with seawater)	Copper-nickel alloy	ASTM B369	C96400
	Aluminum bronze	ASTM B148	C95800
Collar, hand wheel operators (parts not in contact with seawater)	Cast tin bronze	ASTM B61	C92200
	Aluminum bronze	ASTM B505/B505M	C95800
	CRES	ASTM A276/A276M	C31600
Scraper	NiCu	ASTM B127	N04400, N04405
Threaded fasteners	NiCu alloy	ASTM F467	N04400, N04405
		ASTM F468	
O-rings	Fluorocarbon	SAE-AS3209	For all except threaded bosses
		SAE-AS3581	Threaded bosses
Screen	CuNi plate	ASTM B122/B122M	C71500
	NiCu plate	ASTM B127	N04400, N04405

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3.4 Screen. The screen shall be made of one unit. The screen shall be of the perforated plate construction. All seam welds shall be continuous. The total clear/open area of the perforations shall be sufficient to allow a maximum clean pressure drop of 6 psi at the design flow rates as shown in table III. Unless otherwise specified (see 7.1), perforated holes are sized as specified in 7.5.3.

3.4.1 Screen strength test. The strainer screen shall be tested by applying a differential pressure across the screen equal to 100 percent of the specified strainer design pressure for the applicable type and class, using water for a duration of 10 minutes. Differential pressure shall be applied in a rapid manner for 1 minute or less. Screens that distort, stretch, or exhibit other weaknesses shall be rejected.

3.5 Actuation. Each strainer screen shall be fitted with a shaft along its rotational axis, which penetrates the compartment cover through a watertight seal. Outside the housing, the shaft shall be capable of being fitted with either a manual hand wheel (Type I) or an electric motor actuator (Type II). Each manually actuated hand wheel shall be sized and configured so that maximum required operating force is less than 50 pounds. For certain hazard areas, such as installation in a JP-5 pump room, the motor actuator (Type II) shall be explosion-proof.

3.6 Scraper blades. The scraper blades in all strainers size 4 inches and above shall be capable of inspection and adjustment via a removable watertight inspection cover on the strainer body, without having to further disassemble the strainer. Each scraper blade shall be aligned so that the screen can be turned in either direction without being damaged.

3.7 Functional performance. Unless otherwise specified (see 7.1), design flow capacity shall be as specified in table III and design temperature shall be 120 °F.

Table III. Functional performance.

Size	Number of Screens	Flow Rate (gpm)
2	1	100
2½	1	100
3	1	160
2½	2	165
3	2	250
4	2	390
6	2	900
8	2	1500
10	2	2150
12	2	2750
14	4	4250
16	4	5500
20	6	8250
NOTE: Screen perforation of $^{1}/_{32}$ inch.		

3.7.1 Operating cycling tests. The strainer shall be tested for a total of 600 cycles at design pressure. At the completion of the 600 cycles, compliance with the required operating force of 50 pounds or less and zero leakage shall be demonstrated. Upon satisfactory completion of this test, the unit shall be refurbished to new condition before being offered for delivery.

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3.8 Envelope dimensions. Maximum envelope dimensions (inches) shall be in accordance with table IV.

Table IV. Envelope dimensions.

Size	Number of Screens	Manual Actuation			Motor Actuation			Screen Removal Clearance
		Length	Width	Flange-to-Flange	Length	Width	Flange-to-Flange	
2	1	13.00	7.00	11.00	---	---	---	13.50
2½	1	14.00	7.00	12.25	---	---	---	15.50
3	1	15.00	10.00	14.25	20.00	21.50	22.50	17.25
2½	2	21.75	6.25	12.50	---	---	---	15.50
3	2	24.00	8.50	13.75	---	---	---	17.50
4	2	28.50	9.50	16.75	34.00	38.00	16.75	21.00
6	2	35.50	14.00	21.00	41.00	44.50	21.00	27.00
8	2	44.00	19.50	25.00	46.00	47.50	25.00	33.00
10	2	51.50	23.75	31.00	54.00	49.00	31.00	39.00
12	2	56.50	25.50	35.25	61.50	50.00	35.25	44.00
14	4	64.00	64.00	31.75	56.00	63.00	42.00	40.00
16	4	71.50	71.50	34.75	70.00	77.00	45.00	44.00
20	6	76.00	76.00	41.50	68.00	75.00	51.00	45.00

NOTES:

1. Flange-to-flange dimension for single screen strainers are measured from the inlet flange to the flush outlet opening.
2. Width dimension is defined as the widest point of the unit (i.e., 90-degree to hand wheel or motor).
3. Length dimension is defined as the longest point of the unit (i.e., hand wheel to hand wheel).
4. All dimensions are in inches.

3.9 Workmanship. The strainer body and covers shall be of uniform quality and condition, free from blow holes, porosity, hard spots, shrinkage defects, cracks, and other defects. All surfaces shall be smooth and well cleaned. The inside surfaces of strainers shall be well cleaned and free from sharp edges.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Prohibited material. Zinc or cadmium plating shall not be used on any part of the strainer.

4.3 Mercury exclusion. The strainer shall be free of mercury contamination. During the manufacturing process, tests, and examination, the product to be offered for acceptance shall not come in direct contact with mercury or any of its components, nor with any mercury containing device employing a single boundary of containment.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

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5.2 Additional inspection. Unless otherwise specified in the contract or order (see 7.1), where other specifications form a part of this CID, sampling, examination, and tests shall be conducted as required by the pertinent specification.

6. PACKAGING.

6.1 General requirement. Commercial packaging shall be in accordance with the requirements of ASTM D3951.

6.2 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two 3- by 4-inch nominal wood skids laid flat or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage, and storage.

7. NOTES.

7.1 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN (see 7.5).
b. Type, class, style, and size required (see 2.1, 2.2, and 2.3).
c. Strainer screen design and characteristics (see 3.4).
d. The required design flow capacity and temperature if not as specified (see 3.7).
e. Product conformance provisions (see 5.2).
f. Whether repair parts are required (see 7.3.1).

7.2 Source of documents.

7.2.1 ASTM International. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

7.2.2 Defense specifications and standards. Defense specifications and standards are available online at http://quicksearch.dla.mil.

7.2.3 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at https://www.acquisition.gov/far.

7.2.4 Fluid Control Institute (FCI). FCI standards are available from the Fluid Controls Institute, Inc., 1300 Sumner Avenue, Cleveland, OH 44115 or online at http://www.fluidcontrolsinstitute.org/.

7.2.5 International Society of Automation (ISA). ISA standards are available from the International Society of Automation, 67 T.W. Alexander Drive, P.O. Box 12277, Research Triangle Park, NC 27709 or online at www.isa.org.

7.2.6 SAE International. SAE standards are available from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or online at www.sae.org.

7.3 Ordering spare or repair parts. When ordering spare parts or repair parts for the equipment covered by this CID, the contract should state that such spare parts and repair parts should meet the same requirements and provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

7.3.1 Repair parts and repair parts boxes. Complete sets of repair strainer screens for each type and class should be furnished as follows:

a. One set for each of the ten initial strainers required.
b. One set for each two additional strainers required.
c. Cover O-rings - four for every strainer supplied.
d. Other O-rings and seals - one set for every ten strainers supplied.

When specified (see 7.1), 7.3.1.c and 7.3.1.d should be furnished in repair boxes.

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7.4 Preinstallation instruction. A set of instructions covering the preinstallation of the equipment should be furnished. Instructions should include all information necessary to return the unit to active status, such as, but not limited to, the addition of lubricants prior to operation, flushing of lines, removal of greaseproof barrier, and the location of detached components. Instructions should be packaged in a transparent waterproof plastic bag. Closure should be by heat sealing. The shipping container in which the instructions are packed should be so marked.

7.5 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PIN is a definitive number which corresponds to the type, class, style, size, and screen perforation of strainers covered by this CID. The CID number, the type and class code letter, the style and size code numbers, and screen perforation code letter are combined to form a definitive PIN. PINs for the strainers are assigned as follows:

AA59000	**X**	**XX**	**X**
CID Number	Type and Class (see 7.5.1)	Style and Size (see 7.5.2)	Screen Perforation (see 7.5.3)

7.5.1 Type and class. The type and class of strainer (see 2.1 and 2.2) are identified by a single letter (see table V).

Table V. Code letter for type and class.

Class	**Type**	
	Type I	**Type II**
Class 150	A	B
Class 250	C	D

7.5.2 Style and size. The style and size of strainer (see 2.3) are identified by a double digit number (see table VI).

Table VI. Code number for style and size.

Size	**Style**			
	Style 1	**Style 2**	**Style 3**	**Style 4**
2	01	-	-	-
2½	02	04	-	-
3	03	05	-	-
4	-	06	-	-
6	-	07	-	-
8	-	08	-	-
10	-	09	-	-
12	-	10	-	-
14	-	-	11	-
16	-	-	12	-
20	-	-	-	13

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7.5.3 Screen perforation. The screen perforation of strainers is identified by a single letter (see table VII).

Table VII. Code letter for screen perforation.

Size (Inch)	System Particle Size Tolerance Range (Inch)	Code
1/4	0.47 to 0.62	A
3/16	0.31 to 0.47	B
1/8	0.24 to 0.31	C
3/32	0.15 to 0.24	D
1/16	0.09 to 0.15	E
1/32	0.06 to 0.09	F

7.5.4 Example PIN. AA59000C01B is an angled flow, 2-inch single screen (Style 1) strainer with a 250 psig pressure rating (Class 250) and a screen perforation of 3/16 inch fitted with a manual hand wheel actuator (Type I).

7.6 Key words.

Cleaning
Filtration
Motor
Screen
Seawater

MILITARY INTERESTS:

Custodians:
Army – AR
Navy – SH
Air Force – 16

Preparing Activity:
Navy – SH
(Project 4730-2017-031)

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